

Ashton E. Thomas

EDUCATION

University of Michigan - Rackham Graduate School <i>Major - MSE Computer Engineering, Computer Vision Concentration</i> <i>summa cum laude, GPA - 4.00/4.00</i>	08/2025 – 05/2026 <i>Ann Arbor, MI, USA</i>
University of Michigan College of Engineering <i>Major - BSE Computer Science; Minor - International Engineering</i> <i>summa cum laude, GPA - 3.8/4.0</i>	08/2022 – 05/2025 <i>Ann Arbor, MI, USA</i>
<i>Relevant coursework: AR/VR, Computer Organization, Computer Vision, Cryptography, Data Analytics</i> <i>Data Structures & Algorithms, Operating Systems, Quantum Computing, Web Development, Web Systems</i>	

EXPERIENCE

Amazon <i>Software Development Engineer I, AWS</i>	09/2024 – Present <i>Boston, MA & New York, NY, USA</i>
- Build developer-insights platforms at AWS for software delivery health, observability, and engineering analytics as part of a 12+ person team. - Completed an SDE internship on Alexa endpoint experiences, shipping features into Alexa+ with a 20+ engineer team using React Native, Kotlin, and TypeScript. - Managed Amazon company-wide employee data for 100,000s of employees as part of a 10-engineer team, including collection, processing, and executive-level inferences.	
Ground Vehicle Systems Center (SEC) <i>Software Development Intern</i>	05/2024 – 07/2024 <i>Warren, MI, USA</i>
- Leveraged MagicDraw & Excel to design databases for Jira tickets and hardware, leading to improved efficiency. - Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.	
Madi Taylor Photo <i>Full Stack Developer</i>	06/2022 – 07/2024 <i>Traverse City, MI, USA</i>
- Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS. - Implemented robust back-end payment solutions and form validation to streamline user transactions.	

RESEARCH

Polk Lab — University of Michigan <i>Computational Neuroscience Research Assistant</i>	09/2024 – 06/2025 <i>Ann Arbor, MI, USA</i>
- Collaborated with a PhD student to develop a machine learning model aimed at mimicking neural distinctiveness of the human brain, while also enabling the separation of background noise from spoken language. - Worked alongside Prof. Thad Polk to develop a script that processes CSV files with 1000s of columns, utilizing wildcard parsing to dynamically filter columns. The script includes a node hierarchy for efficient column interrelation and supports operations such as adding, removing, printing, and row filtration.	

PROJECTS

Geoguessr AI, Computer Vision	2023 – 2024
- Designed and implemented a modified ResNet-50 architecture for geographic location identification from images. - Fine-tuned the model with 61k images, addressing lighting and seasonal challenges whilst achieving ~90% accuracy.	
Virtual Memory Pager, Operating Systems	2025
- Built a virtual-memory pager managing per-process arenas, page tables, physical frames, swap-backed pages, and shared file-backed mappings. - Implemented page-fault handling, copy-on-write behavior, dirty/reference tracking, swap allocation, file-block sharing, eviction, and a clock-style page replacement policy.	
Invariant Learning Experiments, Machine Learning	2025
- Compared empirical risk minimization, invariant risk minimization, and IRM-Games under distribution shift across controlled experimental environments. - Evaluated model behavior on Colored MNIST, synthetic chain-equation data, and financial time-series settings with spurious correlations, trend regimes, and volatility shifts.	

TECHNICAL SKILLS

ARM, ARCGIS, AWS, Lambda, S3, Glue, Athena, CloudWatch, Redshift, CDK, C#, C++, CI/CD, CSS, Excel, Flask, Git, HTML, Java, Javascript, Jinja, Jira, Keras, Kotlin, Latex, MagicDraw, Matlab, NumPy, Pandas, Python, PyTorch, Qiskit, React/Native, SQL, TensorFlow, Typescript, VSCode
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